

Gravitation – Top 5 Most Critical Exam Questions (2026)

Applicable for: CBSE Class 9 & 11, JEE, and NEET

This document consolidates the **five most frequently asked and conceptually critical questions** from the chapter **Gravitation**. These questions are repeatedly tested in board exams and competitive examinations and should be prepared thoroughly with clear concepts and correct mathematical expressions.

1. State the Universal Law of Gravitation and Derive Its Mathematical Expression

Statement:

Every object in the universe attracts every other object with a force that is: - directly proportional to the product of their masses, and - inversely proportional to the square of the distance between their centres.

Mathematical Expression:

$$F = G \frac{m_1 m_2}{r^2}$$

Where: - F = gravitational force between two objects - m_1, m_2 = masses of the two objects - r = distance between their centres - G = universal gravitational constant

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

2. Establish the Relationship Between Acceleration Due to Gravity (g) and Universal Gravitational Constant (G)

The acceleration due to gravity experienced by an object near the surface of a planet depends on: - the mass of the planet, and - the radius of the planet,

and **not on the mass of the object**.

Derived Formula:

$$g = \frac{GM}{R^2}$$

Where: - g = acceleration due to gravity - M = mass of the planet - R = radius of the planet - G = universal gravitational constant

3. Explain the Difference Between Mass and Weight

Mass	Weight
Quantity of matter in a body	Gravitational force acting on a body
Constant everywhere	Varies with location
SI unit: kilogram (kg)	SI unit: newton (N)
Independent of gravity	Depends on value of g

Important Concept:

A body weighs **one-sixth on the Moon** compared to Earth because the Moon's gravitational pull is much weaker.

4. How Does the Value of g Change with Altitude and Depth?

- The value of acceleration due to gravity **decreases with increase in altitude** above the Earth's surface.
- The value of g also **decreases with increase in depth** below the Earth's surface.

Key Result:

At the **centre of the Earth**, the value of $g = 0$, and an object becomes **effectively weightless**.

5. What Is the Importance of the Universal Law of Gravitation?

The Universal Law of Gravitation explains several natural phenomena, including: - The force that binds objects to the Earth - The motion of the Moon around the Earth - The motion of planets around the Sun - The formation of ocean tides due to the Moon and the Sun

Exam Tip

For full mastery of this chapter, practice: - NCERT textbook questions - NCERT Solutions (step-by-step) - Previous years' CBSE, JEE, and NEET questions

This chapter is **highly conceptual**—clarity in definitions, formulas, and reasoning is essential for scoring well.